Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*
Code: 101303 *Subject: Computer-Aided Civil Engineering Drawing* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (any seven question only):*** **[2 x 7 = 14]**

(a) What does the EXPLODE command do in AutoCAD?
- (i) Breaks a complex object into its constituent parts.
- (ii) Removes an object from a drawing.
- (iii) Detaches an xref
- (iv) Breaks a polyline into individual lines.

(b) How do you create a 3D solid in AutoCAD?
- (i) Using the solid command
- (ii) Using the 3D command
- (iii) Using the EXTRUDE command
- (iv) Both a and c

(c) The right-hand rule is used to determine the direction of the
- (i) Positive X axis.
- (ii) Negative Y axis.
- (iii) Positive Z axis.
- (iv) Negative X axis.

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(d) The ISOLINES system variable specifies
- (i) The number of segments in a tabulated model
- (ii) The number of contour lines that define a solid model.
- (iii) The direction of contour lines in a solid model.
- (iv) The length of the segments in an extrusion.

(e) Which of the following is the correct statement for dimensioning a view?
- (i) All of them should be placed within a view.
- (ii) All of them should be placed close to the parts being dimensioned.
- (iii) They should be marked from visible outlines rather than from hidden line.
- (iv) Dimension indicated in one view may be repeated in another view.

(f) A room of 3025 m^2 area is shown as 25 cm^2 area on a drawing sheet. What is the RF?
- (i) 1/110
- (ii) 1/1100
- (iii) 1/220
- (iv) 1/2200

(g) Scale used for two systems of units measurement is
- (i) Plain scale
- (ii) Diagonal scale
- (iii) Comparative scale
- (iv) Vernier scale

(h) Which line is drawn to make the section evident?
- (i) Long break line
- (ii) Chain thick
- (iii) Border line
- (iv) Hatching line

(i) AutoCAD was first released in the year:
- (i) 1980
- (ii) 1982
- (iii) 1984
- (iv) 1986

(j) How many units are available in AutoCAD?
- (i) 4
- (ii) 5
- (iii) 6
- (iv) 7

Q.2 (a) Draw the detailed plan and section of RCC column and column footing with require details. **[7]**
- (i) Size of footing – 1200 mm × 1200 mm
- (ii) Size of column – 300 mm × 300 mm

(b) Write down the different super structure of building and explain the principles of planning of college building. **[7]**

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Q.3 (a) Give neat sketches and indicate conventional signs for the following: [7]
(i) Railway line (ii) Timber (iii) Steel
(iv) Plain concrete (v) Rock (vi) Plaster
(vii) Brick Masonry

(b) What is the difference between conventional sign & conventional symbol? Explain with neat sketch. [7]

Q.4 (a) Define the followings: [7]
(i) Filler / Extender (ii) Base (iii) Vehicle
(iv) Thinner (v) Drier (vi) Pigment

(b) Differentiate between gravity drainage system & siphonic drainage system [7]

Q.5 (a) What do you understand by squint junction? Draw typical sketches showing squint junction in [7]
(i) English bond (ii) Double Flemish bond

(b) Define the following with net sketch: [7]
(i) Double Flemish bond (ii) English bond
(ii) Racking bond (iv) Burnt clay perforated bricks

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Q.6 (a) It is proposed to construct a residential building (G+1) RCC framed structure with the following requirements: [7]
(i) Drawing hall (ii) Living room
(iii) Kitchen cum dinning (iv) Master bedroom
Provide veranda, passage, staircase, and sanitary unit as per requirement plot size of 50 m × 40 m. Draw floor plan for both levels.

(b) What is BIM? Explain the component of Building Information Modeling (BIM). [7]

Q.7 (a) How are the perspective projections classified? Mention application of each type of perspective projection. [7]

(b) How do you begin a new drawing? How do you select an existing drawing for editing? List five important edit commands and mention their applications. [7]

Q.8 (a) Explain the following: [7]
(i) Reference plane (ii) Layers of AUTOCAD
(ii) Optimal layout of drawing (iv) Picture plane

(b) Explain the key points that must be considered for drawing the foundation plan of load-bearing structures. [7]

Q.9 (a) Explain the various types of line used in civil engineering drawing and write down symbols are used in civil engineering drawing for electrical system? Draw neat sketches of electrical symbols. [7]

(b) What are the benefits of AutoCAD from civil engineering point of view and explain reason for implementing AutoCAD. [7]

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